CHANDIGARH UNIVERSITY
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PHASE III PROJECT EVALUATION REPORT

Faithful or Plausible? Quantitative Evaluation of Explainable AI for 3D MRI-Based Alzheimer's Disease Classification

Feasibility & Budgeting · Features · Constraints · Standards · Design Selection · Execution Plan

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Contents

1. Project Overview 3
2. Feasibility and Budgeting 3
2.1 Technical Feasibility 3
2.2 Data and Legal Feasibility 4
2.3 Computational Feasibility 4
2.4 Schedule Feasibility 4
2.5 Budget 5
3. Project Features and Characteristics 5
3.1 Features 5
3.2 Objectives 6
3.3 Hypotheses 6
4. Identification of Constraints 6
5. Analysis and Finalisation of Features Subject to Constraints 7
5.1 Finalised Experiment Set 8
6. Identification of Appropriate Standards 8
7. Design Selection 9
7.1 Data Source 9
7.2 Atrophy Model 9
7.3 Benchmark Conditions 9
7.4 Classifier and Explanation Methods 10
7.5 System Architecture 10
8. Execution Plan 10
8.1 Risks and Mitigation 11
9. Implementation Status and First Results 11
9.1 Benchmark Results 12
10. References 13

1. Project Overview

Deep-learning classifiers for Alzheimer's disease (AD) from structural MRI are routinely accompanied by explanation heatmaps, and a heatmap on the hippocampus is commonly taken as evidence that the model has learned the disease. This project tests that assumption by separating two properties of an explanation: **plausibility**, whether the map falls on anatomy a clinician expects, and **faithfulness**, whether the highlighted voxels actually drive the prediction.

Phase II planned to study this on the ADNI cohort. Access to ADNI, and subsequently to OASIS, was refused. Phase III therefore presents a redesigned execution plan built on open data: healthy T1-weighted scans from the IXI dataset, on which Alzheimer-like hippocampal atrophy and, in some experiments, a planted shortcut are simulated. Because the disease signal is inserted by us, its location is known voxel by voxel. This adds a third axis, **correctness** against ground truth, which no real-data study can measure, and lets the central claim that plausible maps can be unfaithful be demonstrated rather than argued.

This report covers the Phase III criteria: feasibility and budgeting, project features, constraints, the analysis and finalisation of features under those constraints, applicable standards, design selection, and the execution plan. Section 9 reports the implementation status and the first complete benchmark results: 5-fold cross-validation on all 581 IXI subjects.

Item	Details
Project title	Faithful or Plausible? Quantitative Evaluation of Explainable AI for 3D MRI-Based Alzheimer's Disease Classification
Domain	Explainable AI (XAI), medical image analysis, deep learning
Data (Phase III)	IXI healthy T1 MRI (~580 subjects, CC BY-SA 3.0) with simulated hippocampal atrophy and planted shortcuts; Zenodo AD/BD/HC MRI (CC BY 4.0) as an optional real-data plausibility check
Model	3D ResNet-10 (half width) on 96^3 volumes in MNI space
XAI methods	LRP (composite z^B / α1β0 / ε, BatchNorm folded), Grad-CAM, Grad-CAM++, Integrated Gradients, Occlusion, Gradient × Input
Evaluation axes	Plausibility (atlas regions), faithfulness (insertion / deletion AUC), correctness (agreement with ground-truth maps)
Hardware	Team laptop with NVIDIA RTX 5080 Laptop GPU (16 GB)
Code repository	github.com/UdayBhardwaj/alz
Target output	Research paper (MICCAI workshop such as iMIMIC, then journal) and final-year project report

Table 1. Project at a glance

2. Feasibility and Budgeting

2.1 Technical Feasibility

The framework is implemented and verified on synthetic data before any real scan is used. In its last full run (30 October 2026) the synthetic test suite passed 18 of 18 tests. Three results establish that the core method is implementable correctly:

- **LRP conservation.** Input relevance sums to the model's output logit with a relative error of about 1e-5 on a bias-free network. Before a sign error in the input-layer z^B rule was corrected, the same check gave 1.97, i.e. relevance summed to roughly minus the logit.
- **BatchNorm folding.** The folded network reproduces the original network's outputs to a maximum absolute difference of about 1e-6, tested with randomised BatchNorm statistics so that an incorrect fold cannot pass.
- **Simulation.** On a phantom, a 25% target hippocampal volume loss achieved 22.9%, the deformation never folded (minimum Jacobian determinant 0.885), 92.8% of the ground-truth mass lay within two falloff widths of the structure, and the planted shortcut changed zero voxels outside its ground-truth mask.

2.2 Data and Legal Feasibility

IXI is downloadable without an application and is released under CC BY-SA 3.0, which permits research use with attribution; derived benchmark data must be shared under the same licence. It contains one T1 scan per subject from three London sites, with age and sex. All data are de-identified and public, so no participant contact or new ethical approval is required. The Harvard-Oxford atlas and MNI152 template are distributed with nilearn.

2.3 Computational Feasibility

The classifier is deliberately small. The table below gives the workload on the team laptop as measured in the full 5-fold run.

Workload	Size	Measured
Preprocessing (CPU)	581 scans; N4, affine registration to the ICBM152 head template, template brain mask	0 failures; ~40 s per scan per worker; ~1.5 h with 5 workers
Benchmark generation (CPU)	6 cohort builds × 581 scans	~1 s per scan
Model	3D ResNet-10 half width, 96^3 input	Fits the 16 GB GPU at batch 8, mixed precision
Training	20 runs (4 conditions × 5 folds)	1–3 min each on the GPU
Evaluation	6 methods + random map × 581 subjects × 7 test sets	~20 min per test set
Storage	Raw, preprocessed and benchmark volumes with ground-truth maps	~40 GB

Table 2. Computational workload (measured)

2.4 Schedule Feasibility

The full 5-fold benchmark was completed in October 2026, well ahead of the plan, because training and evaluation proved far faster than estimated. The remaining experiments (full shortcut sweep, randomisation tests, ablations) fit November 2026, leaving December 2026 to March 2027 for writing and submission before the final evaluation in April–May 2027. CPU temperature during parallel preprocessing (up to 98 °C) was the one practical constraint; it was managed by halving the number of parallel workers.

2.5 Budget

The redesign removes the cloud-GPU dependency of the Phase II plan. The committed cost is zero; optional items are capped and require supervisor approval before any spending.

Item	Plan	Cost
GPU compute	Team laptop, RTX 5080 Laptop GPU (16 GB)	₹0 (owned)
CPU preprocessing	Same laptop, parallel ANTs registration	₹0
Data	IXI, Harvard-Oxford atlas, MNI152 template	₹0 (open)
Storage	~40 GB local SSD	₹0 (owned)
Software	Python, PyTorch, ANTsPy, nibabel, nilearn, scikit-learn, Zennit	₹0 (open source)
Cloud GPU (optional)	Hourly GPU only if laptop throttling blocks the schedule; stopped when idle	Capped; approval first
Publication (optional)	Workshop or conference registration if accepted	As per venue
Total committed		**₹0**

Table 3. Budget

3. Project Features and Characteristics

3.1 Features

The system is organised as six features. Each is a separate module with its own tests, so a defect can be traced to one stage.

ID	Feature	Characteristics	Module
F1	Data pipeline	IXI download and demographics; N4 bias correction, skull stripping, affine MNI registration, 96^3 resampling, brain-masked z-scoring; resumable	data/ixi_dataset.py, data/preprocess.py
F2	Ground-truth benchmark	Per-component hippocampal shrink with a voxel-wise $\|\det J - 1\|$ truth map; planted shortcut with exactly controlled label correlation; shared sub-voxel warp for every scan; site- and age-balanced labels	data/simulate.py, data/build_benchmark.py
F3	Leakage-safe classifier	3D ResNet; subject-level folds; split check that raises an error on any shared subject	models/resnet3d.py, data/adni_dataset.py
F4	Six attribution methods	LRP with residual-aware rules and BatchNorm folding; Grad-CAM, Grad-CAM++, Integrated Gradients, Occlusion, Gradient × Input; all target the model's own prediction	xai/
F5	Three-axis evaluation	Plausibility: relevance share in AD atlas regions. Faithfulness: insertion / deletion AUC vs a random map. Correctness: mass fraction, AUPRC and precision@k against ground truth	eval/anatomical.py, eval/faithfulness.py
F6	Verification first	18 synthetic tests; conservation check; cascading randomisation sanity checks; fixed seeds; tests are never weakened to pass	tests/, eval/sanity.py

Table 4. Project features

3.2 Objectives

The Phase II objectives are retained in intent and restated for the new data design.

#	Objective	Measurable outcome
O1	Build an open-data benchmark with voxel-level ground truth	Achieved atrophy volume loss recorded per subject; shortcut changes no voxel outside its mask; controls vs controls at chance
O2	Train a leakage-safe 3D classifier on each condition	Subject-level 5-fold CV; test AUC clearly above chance on condition A; resampling control at chance
O3	Implement six attribution methods correctly	LRP conservation error < 1e-2 (currently ~1e-5); agreement with Zennit < 1e-3
O4	Quantify faithfulness	Insertion / deletion AUC against a random-map baseline, with a perturbation-baseline sweep
O5	Quantify plausibility	Relevance share in medial temporal, posterior cortical and ventricular atlas regions
O6	Quantify correctness against ground truth (new)	Mass fraction, AUPRC (with chance level) and precision@|GT| against atrophy and shortcut maps
O7	Test divergence and robustness	Per-volume rank correlations with subject-level bootstrap CIs; sanity checks; ablations with Kendall's W

Table 5. Objectives

3.3 Hypotheses

The hypotheses are fixed before any benchmark result is examined.

- **H1 – Divergence:** across methods and conditions, plausibility and faithfulness are only weakly correlated per volume ($|\rho| < 0.3$).
- **H2 – Ground truth:** on shortcut-reliant models (conditions B with high p, and C), at least one method keeps high hippocampal plausibility while its correctness against the shortcut map is low: plausible but provably unfaithful.
- **H3 – Resolution:** Grad-CAM, upsampled from a 6^3 feature map, scores relatively higher on plausibility than on correctness; LRP shows the reverse.
- **H4 – Sanity:** methods differ in how fast their maps decorrelate under cascading weight randomisation; a method that stays near $\rho = 1$ fails.
- **H5 – Metric validity (new):** in condition A, the ranking of methods by insertion / deletion faithfulness agrees with their ranking by ground-truth correctness. If it does not, the faithfulness metric itself is unreliable in 3D MRI.

4. Identification of Constraints

Constraints were identified across data, validity, compute, methodology, schedule and licensing. Severity reflects how strongly each one shapes the design.

#	Constraint	Description and impact	Severity
C1	Data access	ADNI and OASIS refused access. No labelled real AD cohort is available to the team, which removes the Phase II main cohort.	Hard

#	Constraint	Description and impact	Severity
C2	External validity	Simulated atrophy is not Alzheimer's disease. Results can support claims about explanation methods, not about diagnosis.	Hard
C3	Compute	One laptop GPU with 16 GB memory, subject to thermal throttling on long runs.	Medium
C4	Site confound	IXI was acquired at three hospitals on different scanners; if labels correlated with site, the model could learn the scanner.	Medium
C5	Resampling artefact	Warping a volume interpolates and smooths it; a model could detect the smoothing instead of the atrophy.	Medium
C6	Subtle signal	15–30% hippocampal volume loss moves the boundary by under one 2 mm voxel; the classifier may struggle to learn it.	Medium
C7	Atlas alignment	Ground truth depends on the hippocampus mask landing on the hippocampus after registration; this is not yet verified on real scans.	Medium
C8	Schedule and licence	Final evaluation April–May 2027. CC BY-SA requires attribution and share-alike for the derived benchmark.	Soft

Table 6. Identified constraints

5. Analysis and Finalisation of Features Subject to Constraints

Each Phase II feature was checked against the constraints above. Features were kept, modified, replaced, added or dropped as follows.

Phase II feature	Constraint	Phase III decision
ADNI CN vs AD cohort	C1	**Replaced** by IXI with simulated atrophy (conditions A–C)
Single planted-shortcut model	–	**Extended** to a sweep over label correlation $p \in \{0.5, 0.7, 0.9, 1.0\}$ plus a flipped-shortcut reliance test
Two evaluation axes	C2	**Added** a third axis, correctness against ground truth
–	C5	**Added** a shared sub-voxel warp for every scan and a controls-vs-controls run that must stay at chance
Age-confound experiment	C1, C4	**Modified**: IXI is healthy only, so labels are balanced within site and across age; site is reported
Test-retest on repeat scans	C1	**Dropped**: IXI has one scan per subject
24 GB cloud GPU	C3	**Replaced** by the team laptop with mixed precision and small batches
MedicalNet pretraining	C3	**Optional**: tried on the pilot fold only if training from scratch underfits
Fixed atrophy severity	C6	**Modified**: severity range 15–30%, raised only if the pilot fold fails, and reported
Six methods, Zennit check, sanity checks	–	**Kept** unchanged

Table 7. Feature analysis against constraints

5.1 Finalised Experiment Set

The finalised experiments map directly to the objectives and hypotheses.

#	Experiment	Answers	Output
E1	Classifier on conditions A, B, C; subject-level 5-fold CV	Is the signal learned?	Classification table
E2	Resampling control: controls vs controls, same warp	Is interpolation a cue?	Chance-level check
E3	Six methods × three axes on condition A	O4–O6, H3, H5	Method comparison tables
E4	Shortcut sweep (p) and flipped-shortcut test sets	H2	Reliance and correctness table
E5	Divergence analysis across methods and conditions	H1	Scatter plots, correlations
E6	Cascading and data randomisation	H4	Sanity curves
E7	Ablations: patch size {4, 8, 16}, baseline, LRP split, severity	Robustness	Kendall's W
E8	Conservation, BatchNorm folding, Zennit agreement	O3	Validity table
E9	Optional: plausibility on real Zenodo AD scans	External sanity	Qualitative figure

Table 8. Finalised experiments

6. Identification of Appropriate Standards

The following standards govern reporting, requirements, data handling, licensing and software engineering.

Standard	Area	How it is applied
CLAIM 2024 (Checklist for AI in Medical Imaging)	Reporting	Paper reports data sources, preprocessing, splits, model, training and evaluation item by item
TRIPOD+AI (2024)	Reporting	Transparent description of the prediction model, its development and validation
ISO/IEC/IEEE 29148:2018	Requirements	Functional and non-functional requirements specified and traced to tests (Phase II FR1–FR10, NFR1–NFR7)
NIfTI-1 file format	Data	All volumes read and written as NIfTI; preprocessed arrays saved as .npy on a fixed grid
MNI152 space; Harvard-Oxford atlas	Data	Common coordinate space for registration, atlas regions and ground-truth masks
CC BY-SA 3.0 (IXI licence)	Licensing	IXI acknowledged in all outputs; derived benchmark released under the same licence
PEP 8, type hints, docstrings	Software	Consistent, reviewable code across all modules
Reproducibility practice	Software	Fixed and recorded seeds; saved split and benchmark manifests; tagged release; tests never weakened to pass

Table 9. Applicable standards

7. Design Selection

7.1 Data Source

Five data options were scored on the four properties the study needs. ✓ = yes, ✗ = no, ~ = partial.

Option	Accessible	Real AD signal	Voxel ground truth	Leakage-safe	Verdict
ADNI	✗	✓	✗	✓	Refused
OASIS-1 / OASIS-3	✗	✓	✗	✓	Refused
Kaggle AD 2D slices	✓	~	✗	✗	Rejected: subjects span splits
Zenodo AD/BD/HC	✓	✓	✗	✓	Optional real-data check
IXI + simulated atrophy	✓	~	✓	✓	**Selected**

Table 10. Data source comparison

IXI with simulation is the only option that is both available and provides ground truth. Because the study's claims are about explanation methods, ground truth is the property that matters most; the partial disease realism is stated as a limitation (C2).

7.2 Atrophy Model

Four ways of introducing an Alzheimer-like signal were considered.

Approach	Strength	Weakness
Intensity change in a cube	Trivial; exact ground truth	Unrealistic; not atrophy
Biophysical simulator (Khanal et al. 2017)	Realistic, physically based	Heavy C++ / PETSc dependency
Generative model (e.g. latent diffusion)	Realistic texture	No voxel ground truth; needs training
Smooth displacement field (selected)	Dependency-free; exact $\lvert \det J - 1 \rvert$ truth; folding checked; achieved loss measured	Phenomenological, not biophysical

Table 11. Atrophy model alternatives

The selected field shrinks each hippocampus uniformly toward its own centroid and fades the boundary displacement into surrounding tissue with a Gaussian of distance. Two earlier variants were rejected during implementation because tests exposed them: a blurred-mask weight under-shrank the structure (13.9% for a 25% target), and a centroid-relative field spread the deformation across the brain (only 61% of ground-truth mass near the structure).

7.3 Benchmark Conditions

Condition	Label-1 scans contain	A correct explanation should highlight
A – Atrophy	Shrunken hippocampi (15–30% volume loss)	Hippocampus

Condition	Label-1 scans contain	A correct explanation should highlight
B – Atrophy + shortcut	Atrophy; a faint intensity bump in a fraction p of label-1 and 1 − p of label-0 scans	Whatever the model relies on, measured by the reliance test
C – Shortcut only	A faint bump far from the hippocampus	The bump, not the hippocampus

Table 12. Benchmark conditions

For each shortcut condition a second test set is built with the shortcut moved to the opposite class. A model that relies on atrophy keeps its accuracy; a model that relies on the shortcut collapses. This measures reliance directly, independent of any explanation method.

7.4 Classifier and Explanation Methods

A 3D network was chosen over 2D slices because atrophy is a volumetric change and 2D slice datasets are prone to subject leakage. ResNet-10 at half width was chosen over deeper networks and vision transformers because the cohort is a few hundred subjects. The six attribution methods are retained from Phase II so that results remain comparable. Insertion / deletion AUC was chosen as the faithfulness metric over ROAR, whose retraining is prohibitive in 3D; ROAD remains a possible extension.

7.5 System Architecture

#	Stage	Input → Output	Module
1	Acquire	IXI archive → NIfTI scans + demographics	data/ixi_dataset.py
2	Preprocess	NIfTI → 96^3 normalised volume in MNI space	data/preprocess.py
3	Simulate	Volume + hippocampus mask → benchmark volume, atrophy and shortcut truth maps, manifest	data/simulate.py, data/build_benchmark.py
4	Train	Manifest → per-fold checkpoints with split checks	models/resnet3d.py, train.py
5	Explain	Checkpoint + volume → six attribution maps	xai/
6	Evaluate	Maps → plausibility, faithfulness, correctness, sanity	eval/, run_evaluation.py

Table 13. Pipeline stages

8. Execution Plan

Each stage has a gate that must pass before the next begins.

Period	Activities	Gate
Oct 2026	IXI download; preprocessing of all 581 scans; hippocampus-mask QC on scans from two sites	Done: mask on the hippocampus
Oct 2026	Conditions A, B, C and resampling control; train.py; pilot fold; 5-fold training	Done: control AUC 0.515
Oct 2026	Six methods on all 581 subjects; three-axis evaluation; location control	Done: Section 9

Period	Activities	Gate
Nov 2026	Shortcut sweep (p = 0.7, 1.0, stronger bump); randomisation tests; patch-size ablation; Zennit cross-check	All numbers generated by scripts
Dec 2026–Mar 2027	Figures, manuscript, CLAIM checklist; submission	Supervisor review
Apr–May 2027	Final report, demonstration video, viva	Phase IV evaluation

Table 14. Execution timeline (status on 30 October 2026)

8.1 Risks and Mitigation

Risk	Mitigation	Status
Hippocampus mask misaligned on real scans	Visual check on preprocessed scans from two sites before any cohort is built	Checked
Model detects resampling instead of atrophy	Every scan passes through the same warp; the control run must give chance accuracy	Checked: AUC 0.515
Model fails to learn subtle atrophy	Pilot fold; raise severity within a reported range only if needed	Resolved: AUC 0.992
Site becomes an unplanned shortcut	Labels balanced within site and by age; site reported with demographics	Balanced by design
Fixed shortcut position biases ground-truth scores	Location control: a model that never saw a shortcut, scored at the shortcut's position	Added (E9)
Simulated data questioned by reviewers	Frame the work as a benchmark for XAI methods; optional plausibility check on real Zenodo AD scans	Open
Laptop CPU heat during preprocessing	Fewer parallel workers; training and evaluation on the GPU	Managed

Table 15. Risk register

9. Implementation Status and First Results

Status of the codebase on 30 October 2026.

Component	Content	Status
Attribution methods	LRP (z^B sign fixed, BatchNorm folded), Grad-CAM, Grad-CAM++, IG, Occlusion, Gradient × Input	Done, tested
LRP diagnostics	Unnormalised propagation; conservation error	Done, tested
Faithfulness, plausibility, sanity	Insertion / deletion AUC, atlas regions, cascading randomisation	Done, tested
Correctness metrics	Mass fraction, AUPRC, precision@k against ground truth	Done, tested
IXI loader	Download, filename parsing, demographics	Done, tested
Simulation and benchmark builder	Atrophy, shortcut, shared warp, label and shortcut assignment; three conditions and flip sets	Done, tested on phantoms
Preprocessing	N4, head-to-head affine registration to ICBM152, template brain mask, 96^3	Done: 581 of 581 scans

Component	Content	Status
Training script	Subject-level CV, mixed precision, early stopping on AUC	Done: 4 conditions × 5 folds
Evaluation and analysis	Six methods on every test subject; bootstrap CIs over subjects; location control	Done
Shortcut sweep, randomisation tests, ablations, Zennit cross-check	p = 0.7 / 1.0; cascading randomisation; patch size; reference agreement	Planned (Nov)

Table 16. Implementation status

Key implementation finding. The conservation test exposed a sign error in the LRP input rule: bound terms were subtracted although their gradients already carry the minus sign. Relevance therefore summed to roughly minus the logit, yet the heatmaps still looked plausible. After the fix, LRP's faithfulness score on the synthetic model changed from −0.014 (below the random baseline) to +0.035 (above it). This is the project's thesis in miniature: a plausible-looking map is not evidence of a correct one.

9.1 Benchmark Results

The benchmark was run with subject-level 5-fold cross-validation on all 581 IXI subjects; every subject is tested once and explained by all six methods. All numbers below are generated from the analysis output; the research article draft gives the full tables with confidence intervals.

Condition	Test AUC (mean ± sd)	Accuracy	Accuracy, shortcut moved	Logit shift from shortcut
Resampling control	0.515 ± 0.036	0.484	–	–
A: atrophy only	0.992 ± 0.006	0.943	–	–
C: shortcut only	0.826 ± 0.070	0.747	0.253	6.46
B: atrophy + shortcut, p = 0.9	0.980 ± 0.025	0.924	0.921	0.10

Table 17. Classification and shortcut reliance (581 subjects, 5 folds)

Method (condition A)	Faithfulness	Plausibility (× chance)	Correctness (× chance)	Faithfulness, control model
Occlusion	0.859	6.09	3.17	0.695
LRP	0.617	2.14	1.82	0.465
Integrated Gradients	0.558	1.40	1.58	0.408
Gradient × Input	0.546	1.27	1.57	0.400
Grad-CAM	0.301	1.14	1.32	0.053
Grad-CAM++	0.292	1.14	1.25	0.072
Random map	0.013	1.00	1.00	−0.006

Table 18. Three axes on atrophy, and faithfulness on the control model that learned nothing (means over 581 subjects)

The benchmark works as designed: the control model stays at chance, the atrophy model learns the simulated disease, and the flip test separates a model that relies on the shortcut (condition C) from one that barely uses it (condition B). Four findings follow:

- **Faithfulness agrees with correctness.** On atrophy, the six methods rank in the same order on both axes (Spearman ρ = 1.0).
- **Plausibility misleads.** It exaggerates differences between methods (Occlusion 6.1 times chance on the hippocampus against correctness of 3.2), is at most moderately correlated with faithfulness per volume, and its correlation changes sign between conditions.
- **Faithfulness alone is not enough.** On the control model, which learned nothing, LRP still scored 0.47 against 0.62 on the atrophy model.
- **Grad-CAM has a location bias.** Grad-CAM and Grad-CAM++ place 2.7–3.0 times chance relevance at the shortcut's position in a model that never saw a shortcut, enough to look like shortcut detection. LRP, Integrated Gradients and Gradient × Input attribute relevance to the shortcut in proportion to the model's actual dependence on it.

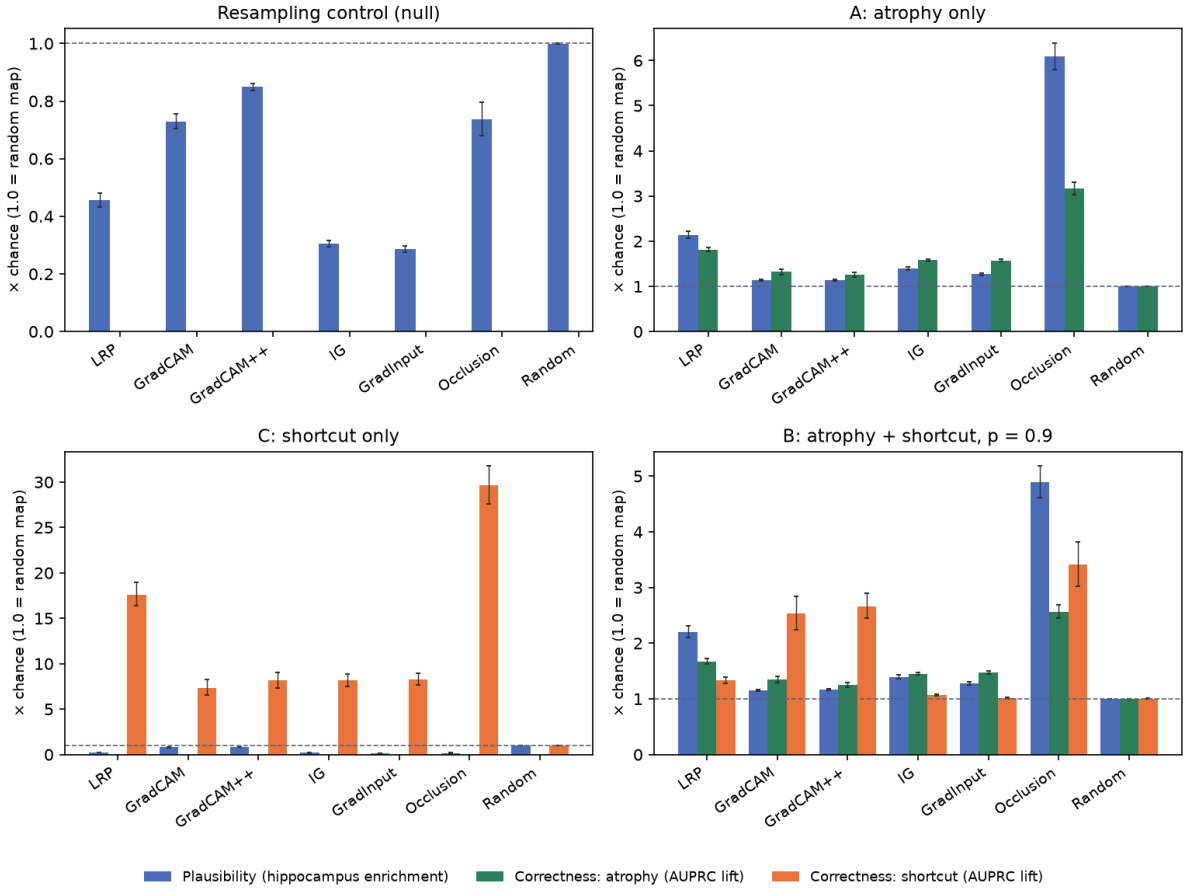


Figure 1. Plausibility and correctness per method and condition, relative to a random map (dashed line at 1.0); error bars are 95% bootstrap CIs over 581 subjects.

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